CSA1017 – SOFTWARE ENGINEERING
INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: K R GAYANA
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ROLE: Project Coordination, Problem Formulation and Scrum Planning

DESIGN / DEVELOPMENT DECISION: I contributed to project coordination, problem formulation, system scope definition, product backlog creation, sprint planning, prioritisation, and integration of the technical modules. I decided to follow the Scrum methodology to divide the development into manageable sprints. MoSCoW prioritisation was used to identify essential features such as early warnings, risk classification, alerts, and authority visibility.

CHALLENGE FACED AND SOLUTION: The major challenge was balancing an ambitious real-world disaster-management system with the limitations of an academic prototype. I addressed this by clearly distinguishing implemented features, simulated data, assumptions, and future improvements. This helped ensure that prototype results were not presented as official disaster forecasts.

LEARNING GAINED: I learned that project coordination involves more than task distribution. It requires maintaining traceability between requirements, design, implementation, testing, and final evidence. I gained practical knowledge of Scrum boards, sprint reviews, acceptance criteria, risk registers, GitHub collaboration, and system integration.

SDG RELEVANCE: The project supports SDG 11 by promoting safer, more resilient, and better-prepared communities. Improved warning communication and evacuation coordination can help reduce confusion during flood emergencies.

COURSE OUTCOME CONTRIBUTION: My contribution supports course outcomes related to requirements engineering, software project management, system design, teamwork, and professional documentation. The work improved my ability to coordinate a technical project while considering safety and system limitations.

POSSIBLE IMPROVEMENT: I would improve the project by incorporating approved live data feeds, field validation, stronger cybersecurity controls, high-availability infrastructure, and official emergency operating procedures.